

- [About](#)
- [Machii](#)
- [Press](#)

note: I haven't had time to write this report since I left the venture world for the startup world, so please treat the below as more of a museum of what machine intelligence was from 2014-2016 😊

The Current State of Machine Intelligence 3.0

(originally published by O'Reilly here, this year in collaboration with my amazing partner James Cham! If you're interested in enterprise implications of this chart please refer to Harvard Business Review's The Competitive Landscape for Machine Intelligence)

Almost a year ago, we published our now-annual landscape of machine intelligence companies, and goodness have we seen a lot of activity since then. This year's landscape has a third more companies than our first one did two years ago, and it feels even more futile to try to be comprehensive, since this just scratches the surface of all of the activity out there.

As has been the case for the last couple of years, our fund still obsesses over "problem first" machine intelligence—we've invested in 35 machine intelligence companies solving 35 meaningful problems in areas from security to recruiting to software development. (Our fund focuses on the future of work, so there are some machine intelligence domains where we invest more than others.)

At the same time, the hype around machine intelligence methods continues to grow: the words "deep learning" now equally represent a series of meaningful breakthroughs (wonderful) but also a hyped phrase like "big data" (not so good!). We care about whether a founder uses the right method to solve a problem, not the fanciest one. We favor those who apply technology thoughtfully.

What's the biggest change in the last year? We are getting inbound inquiries from a different mix of people. For [v1.0](#), we heard almost exclusively from founders and academics. Then came a healthy mix of investors, both private and public. Now overwhelmingly we have heard from existing companies trying to figure out how to transform their businesses using machine intelligence.

For the first time, a "one stop shop" of the machine intelligence stack is coming into view—even if it's a year or two off from being neatly formalized. The maturing of that stack might explain why more established companies are more focused on building legitimate machine intelligence capabilities. Anyone who has their wits about them is still going to be making initial build-and-buy decisions, so we figured an early attempt at laying out these technologies is better than no attempt.

Ready player world

Many of the most impressive looking feats we've seen have been in the gaming world, from DeepMind beating Atari classics and the world's best at Go, to the OpenAI gym, which allows anyone to train intelligent agents across an array of gaming environments.

The gaming world offers a perfect place to start machine intelligence work (e.g., constrained environments, explicit rewards, easy-to-compare results, looks impressive)—especially for reinforcement learning. And it is much easier to have a self-driving car agent go a trillion miles in a simulated environment than on actual roads. Now we're seeing the techniques used to conquer the gaming world moving to the real world. A newsworthy example of game-tested technology entering the real world was when DeepMind used neural networks to make Google's data centers more efficient. This begs questions: What else in the world looks like a game? Or what else in the world can we reconfigure to make it look more like a game?

Early attempts are intriguing. Developers are dodging meter maids (brilliant—a modern day Paper Boy), categorizing cucumbers, sorting trash, and recreating the memories of loved ones as conversational bots. Otto's self-driving trucks delivering beer on their first commercial ride even seems like a bonus level from Grand Theft Auto. We're excited to see what new creative applications come in the next year.

Why even bot-her?

Ah, the great chatbot explosion of 2016, for better or worse—we liken it to the mobile app explosion we saw with the launch of iOS and Android. The dominant platforms (in the machine intelligence case, Facebook, Slack, Kik) race to get developers to build on their platforms. That means we'll get some excellent bots but also many terrible ones—the joys of public experimentation.

The danger here, unlike the mobile app explosion (where we lacked expectations for what these widgets could actually do), is that we assume anything with a conversation interface will converse with us at near-human level. Most do not. This is going to lead to disillusionment over the course of the next year but it will clean itself up fairly quickly thereafter.

When our fund looks at this emerging field, we divide each technology into two components: the conversational interface itself and the "agent" behind the scenes that's learning from data and transacting on a user's behalf. While you certainly can't drop the ball on the interface, we spend almost all our time thinking about that behind-the-scenes agent and whether it is actually solving a meaningful problem.

We get a lot of questions about whether there will be "one bot to rule them all." To be honest, as with many areas at our fund, we disagree on this. We certainly believe there will not be one agent to rule them all, even if there is one interface to rule them all. For the time being, bots will be idiot savants: stellar for very specific applications.

We've written a bit about this, and the framework we use to think about how agents will evolve is a CEO and her support staff. Many Fortune 500 CEOs employ a scheduler, handler, a research team, a copy editor, a speechwriter, a personal shopper, a driver, and a professional coach. Each of these people performs a dramatically different function and has access to very different data to do their job. The bot / agent ecosystem will have a similar separation of responsibilities with very clear winners, and they will divide fairly cleanly along these lines. (Note that some CEO's have a chief of staff who coordinates among all these functions, so perhaps we

will see examples of “one interface to rule them all.”)

You can also see, in our landscape, some of the corporate functions machine intelligence will re-invent (most often in interfaces other than conversational bots).

On to **11111000001**

Successful use of machine intelligence at a large organization is surprisingly binary, like flipping a stubborn light switch. It’s hard to do, but once machine intelligence is enabled, an organization sees everything through the lens of its potential. Organizations like Google, Facebook, Apple, Microsoft, Amazon, Uber, and Bloomberg (our sole investor) bet heavily on machine intelligence and have its capabilities pervasive throughout all of their products.

Other companies are struggling to figure out what to do, as many boardrooms did on “what to do about the Internet” in 1997. Why is this so difficult for companies to wrap their heads around? Machine intelligence is different from traditional software. Unlike with big data, where you could buy a new capability, machine intelligence depends on deeper organizational and process changes. Companies need to decide whether they will trust machine intelligence analysis for one-off decisions or if they will embed often-inscrutable machine intelligence models in core processes. Teams need to figure out how to test newfound capabilities, and applications need to change so they offer more than a system of record; they also need to coach employees and learn from the data they enter.

Unlike traditional hard-coded software, machine intelligence gives only probabilistic outputs. We want to ask machine intelligence to make subjective decisions based on imperfect information (eerily like what we trust our colleagues to do?). As a result, this new machine intelligence software will make mistakes, just like we do, and we’ll need to be thoughtful about when to trust it and when not to.

The idea of this new machine trust is daunting and makes machine intelligence harder to adopt than traditional software. We’ve had a few people tell us that the biggest predictor of whether a company will successfully adopt machine intelligence is whether they have a C-Suite executive with an advanced math degree. These executives understand it isn’t magic—it is just (hard) math.

Machine intelligence business models are going to be different from licensed and subscription software, but we don’t know how. Unlike traditional software, we still lack frameworks for management to decide where to deploy machine intelligence. Economists like Ajay Agrawal, Joshua Gans, and Avi Goldfarb have taken the first steps toward helping managers understand the economics of machine intelligence and predict where it will be most effective. But there is still a lot of work to be done.

In the next few years, the danger here isn’t what we see in dystopian sci-fi movies. The real danger of machine intelligence is that executives will make bad decisions about what machine intelligence capabilities to build.

Peter Pan’s never-never land

We’ve been wondering about the path to grow into a large machine intelligence company. Unsurprisingly, there have been many machine intelligence acquisitions (Nervana by Intel, Magic Pony by Twitter, Turi by Apple, Metamind by Salesforce, Otto by Uber, Cruise by GM, SalesPredict by Ebay, Viv by Samsung). Many of these happened fairly early in a company’s life and at quite a high price. Why is that?

Established companies struggle to understand machine intelligence technology, so it’s painful to sell to them, and the market for buyers who can use this technology in a self-service way is small. Then, if you do understand how this technology can supercharge your organization, you realize it’s so valuable that you want to hoard it. Businesses are saying to machine intelligence companies, “forget you selling this technology to others, I’m going to buy the whole thing.”

This absence of a market today makes it difficult for a machine intelligence startup, especially horizontal technology providers, to “grow up”—hence the Peter Pans. Companies we see successfully entering a long-term trajectory can package their technology as a new problem-specific application for enterprise or simply transform an industry themselves as a new entrant (love this). We flagged a few of the industry categories where we believe startups might “go the distance” in this year’s landscape.

Inspirational machine intelligence

Once we do figure it out, machine intelligence can solve much more interesting problems than traditional software. We’re thrilled to see so many smart people applying machine intelligence for good.

Established players like Conservation Metrics and Vulcan Conservation have been using deep learning to protect endangered animal species; the ever-inspiring team at Thorn is constantly coming up with creative algorithmic techniques to protect our children from online exploitation. The philanthropic arms of the tech titans joined in, enabling nonprofits with free storage, compute, and even developer time. Google partnered with nonprofits to found Global Fishing Watch to detect illegal fishing activity using satellite data in near real time, satellite intelligence startup Orbital Insight (in which we are investors) partnered with Global Forest Watch to detect illegal logging and other causes of global forest degradation. Startups are getting into the action, too. The Creative Destruction Lab machine intelligence accelerator (with whom we work closely) has companies working on problems like earlier diseasedetection and injury prevention. One area where we have seen some activity but would love to see more is machine intelligence to assist the elderly.

In talking to many people using machine intelligence for good, they all cite the critical role of open source technologies. In the last year, we’ve seen the launch of OpenAI, which offers everyone access to world class research and environments, and better and better releases of TensorFlow and Keras. Non-profits are always trying to do more with less, and machine intelligence has allowed them to extend the scope of their missions without extending budget. Algorithms allow non-profits to inexpensively scale what would not be affordable to do with people.

We also saw growth in universities and corporate think tanks, where new centers like USC’s Center for AI in Society, Berkeley’s Center for Human Compatible AI, and the multiple-corporation Partnership on AI study the ways in which machine intelligence can help humanity. The White House even got into the act: after a series of workshops around the U.S., they published a 48-page report outlining their recommendations for applying machine intelligence to safely and fairly address broad social problems.

On a lighter note, we’ve also heard whispers of more artisanal versions of machine intelligence. Folks are doing things like using computer vision algorithms to help them choose the best cocoa beans for high-grade chocolate, write poetry, cook steaks, and generate musicals.

Curious minds want to know. If you’re working on a unique or important application of machine intelligence we’d love to hear from you.

Looking forward

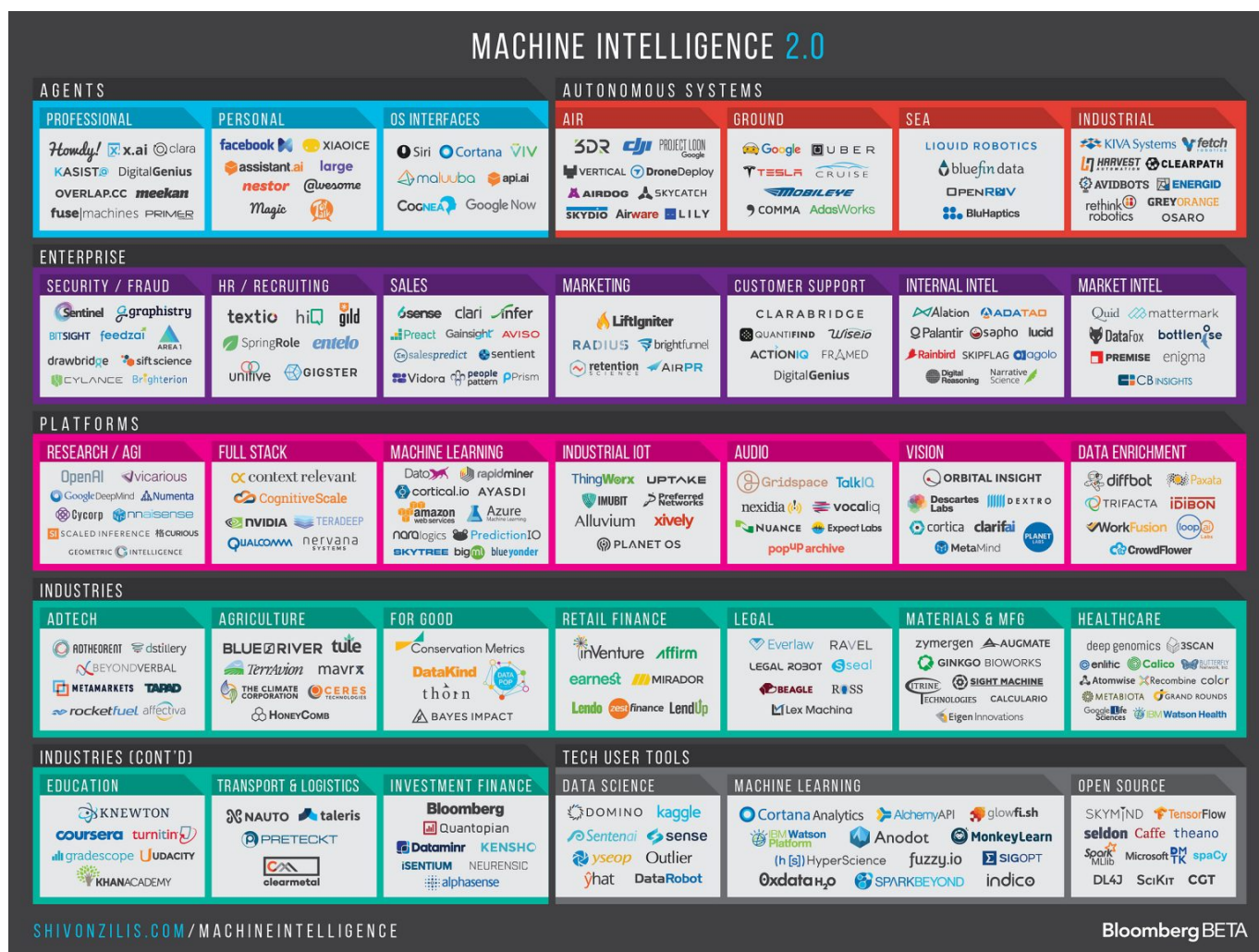
We see all this activity only continuing to accelerate. The world will give us more open sourced and commercially available machine intelligence building blocks, there will be more data, there will be more people interested in learning these methods, and there will always be problems worth solving. We still need ways of explaining the difference between machine intelligence and traditional software, and we're working on that. The value of code is different from data, but what about the value of the model that code improves based on that data?

Once we understand machine intelligence deeply, we might look back on the era of traditional software and think it was just a prologue to what's happening now. We look forward to seeing what the next year brings.

Landscape designed by Heidi Skinner.

Disclosure: Bloomberg Beta is an investor in Alation, Arimo, Aviso, Brightfunnel, Context Relevant, Deep Genomics, Diffbot, Digital Genius, Domino Data Labs, Drawbridge, Gigster, Gradescope, Graphistry, Gridspace, Howdy, Kaggle, Kindred.ai, Mavrx, Motiva, PopUpArchive, Primer, Sapho, Shield.AI, Textio, and Tule.

The Current State of Machine Intelligence 2.0



A year ago, I published my original attempt at mapping the machine intelligence ecosystem. So much has happened since. I spent the last 12 months geeking out on every company and nibble of information I can find, chatting with hundreds of academics, entrepreneurs, and investors about machine intelligence. This year, given the explosion of activity, my focus is on highlighting areas of innovation, rather than on trying to be comprehensive. Figure 1 showcases the new landscape of machine intelligence as we enter 2016:

Despite the noisy hype, which sometimes distracts, machine intelligence is already being used in several valuable ways. Machine intelligence already helps us get the important business information we need more quickly, monitors critical systems, feeds our population more efficiently, reduces the cost of health care, detects disease earlier, and so on.

The two biggest changes I've noted since I did this analysis last year are (1) the emergence of autonomous systems in both the physical and virtual world and (2) startups shifting away from building broad technology platforms to focusing on solving specific business problems.

Reflections on the landscape

With the focus moving from "machine intelligence as magic box" to delivering real value immediately, there are more ways to bring a machine intelligence startup to

market. (There are as many ways to go to market as there are business problems to solve. I lay out many of the options here.) Most of these machine intelligence startups take well-worn machine intelligence techniques, some more than a decade old, and apply them to new data sets and workflows. It's still true that big companies, with their massive data sets and contact with their customers, have inherent advantages—though startups are finding a way to enter.

Achieving autonomy

In last year's roundup, the focus was almost exclusively on machine intelligence in the virtual world. This time we're seeing it in the physical world, in the many flavors of autonomous systems: self-driving cars, autopilot drones, robots that can perform dynamic tasks without every action being hard coded. It's still very early days—most of these systems are just barely useful, though we expect that to change quickly.

These physical systems are emerging because they meld many now-maturing research avenues in machine intelligence. Computer vision, the combination of deep learning and reinforcement learning, natural language interfaces, and question-answering systems are all building blocks to make a physical system autonomous and interactive. Building these autonomous systems today is as much about integrating these methods as inventing new ones.

The new (in)human touch

The virtual world is becoming more autonomous, too. Virtual agents, sometimes called bots, use conversational interfaces (think of *Her*, without the charm). Some of these virtual agents are entirely automated, others are a “human-in-the-loop” system, where algorithms take “machine-like” subtasks and a human adds creativity or execution. (In some, the human is training the bot *while* she or he works.) The user interacts with the system by either typing in natural language or speaking, and the agent responds in kind.

These services sometimes give customers confusing experiences, like mine the other day when I needed to contact customer service about my cell phone. I didn't want to talk to anyone, so I opted for online chat. It was the most “human” customer service experience of my life, so weirdly perfect I found myself wondering whether I was chatting with a person, a bot, or some hybrid. Then I wondered if it even mattered. I had a fantastic experience and my issue was resolved. I felt gratitude to whatever it was on the other end, even if it was a bot.

On one hand, these agents can act utterly professional, helping us with customer support, research, project management, scheduling, and e-commerce transactions. On the other hand, they can be quite personal and maybe we *are* getting closer to *Her*—with Microsoft's romantic chatbot Xiaoice, automated emotional support is already here.

As these technologies warm up, they could transform new areas like education, psychiatry, and elder care, working alongside human beings to close the gap in care for students, patients, and the elderly.

50 shades of grey markets

At least I make myself laugh. 😊

Many machine intelligence technologies will transform the business world by starting in regulatory grey areas. On the short list: health care (automated diagnostics, early disease detection based on genomics, algorithmic drug discovery); agriculture (sensor- and vision-based intelligence systems, autonomous farming vehicles); transportation and logistics (self-driving cars, drone systems, sensor-based fleet management); and financial services (advanced credit decisioning).

To overcome the difficulties of entering grey markets, we're seeing some unusual strategies:

Startups are making a global arbitrage (e.g., health care companies going to market in emerging markets, drone companies experimenting in the least regulated countries).

The “fly under the radar” strategy. Some startups are being very careful to stay on the safest side of the grey area, keep a low profile, and avoid the regulatory discussion as long as possible.

Big companies like Google, Apple, and IBM are seeking out these opportunities because they have the resources to be patient and are the most likely to be able to effect regulatory change—their ability to affect regulation is one of their advantages.

Startups are considering beefing up funding earlier than they would have, to fight inevitable legal battles and face regulatory hurdles sooner.

What's your (business) problem?

A year ago, enterprises were struggling to make heads or tails of machine intelligence services (some of the most confusing were in the “platform” section of this landscape). When I spoke to potential enterprise customers, I often heard things like, “these companies are trying to sell me snake oil” or, “they can't even explain to me what they do.”

The corporates wanted to know what *current* business problems these technologies could solve. They didn't care about the technology itself. The machine intelligence companies, on the other hand, just wanted to talk about their algorithms and how their platform *could* solve hundreds of problems (this was often true, but that's not the point!).

Two things have happened that are helping to create a more productive middle ground:

Enterprises have invested heavily in becoming “machine intelligence literate.” I've had roughly 100 companies reach out to get perspective on how they should think about machine intelligence. Their questions have been thoughtful, they've been changing their organizations to make use of these new technologies, and many different roles across the organization care about this topic (from CEOs to technical leads to product managers).

Many machine intelligence companies have figured out that they need to speak the language of solving a business problem. They are packaging solutions to specific business problems as separate products and branding them that way. They often work alongside a company to create a unique solution instead of just selling the technology itself, being one part educator and one part executor. Once businesses learn what new questions can be answered with machine intelligence, these startups may make a more traditional technology sale.

The great verticalization

I remember reading *Who Says Elephants Can't Dance* and being blown away by the ability of a technology icon like IBM to risk it all. (This was one of the reasons I went to work for them out of college.) Now IBM seems poised to try another risk-it-all transformation—moving from a horizontal technology provider to directly transforming a vertical. And why shouldn't Watson try to be a doctor or a concierge? It's a brave attempt.

It's not just IBM: you could probably make an entire machine intelligence landscape just of Google projects. (If anyone takes a stab, I'd love to see it!)

Your money is nice, but tell me more about your data

In the machine intelligence world, founders are selling their companies, as I suggested last year—but it's about more than just money. I've heard from founders that they are only interested in an acquisition if the acquirer has the right *data set* to make their product work. We're hearing things like, "I'm not taking conversations but, given our product, if X came calling it'd be hard to turn down." "X" is most often Slack (!), Google, Facebook, Twitter in these conversations—the companies that have the data.

(Eh)-I

Until recently, there's been one secret in machine intelligence talent: Canada! During the "AI winter," when this technology fell out of favor in the 80s and 90s, the Canadian government was one of a few entities funding AI research. This support sustained the formidable trio of Geoffrey Hinton, Yoshua Bengio, and Yann LeCun, the godfathers of deep learning.

Canada continues to be central to the machine intelligence frontier. As an unapologetically proud Canadian, it's been a pleasure to work with groups like AICML to commercialize advanced research, the Machine Learning Creative Destruction Lab to support startups, and to bring the machine intelligence world together at events like this one.

So what now?

Machine intelligence is even more of a story than last year, in large companies as well as startups. In the next year, the practical side of these technologies will flourish. Most new entrants will avoid generic technology solutions, and instead have a specific business purpose to which to put machine intelligence.

I can't wait to see more combinations of the practical and eccentric. A few years ago, a company like Orbital Insight would have seemed farfetched—wait, you're going to use satellites and computer vision algorithms to tell me what the construction growth rate is in China!?—and now it feels familiar.

Similarly, researchers are doing things that make us stop and say, "Wait, really?" They are tackling important problems we may not have imagined were possible, like creating fairy godmother drones to help the elderly, computer vision that detects the subtle signs of PTSD, autonomous surgical robots that remove cancerous lesions, and fixing airplane WiFi (just kidding, not even machine intelligence can do that).

Overall, agents will become more eloquent, autonomous systems more pervasive, machine intelligence more...intelligent. I expect more magic in the years to come.

Disclosure: Bloomberg Beta is an investor in Alation, Adatao, Aviso, BrightFunnel, Context Relevant, Deep Genomics, Diffbot, Domino Data Lab, Gigster, Graphistry, Howdy, Kaggle, Mavrx, Orbital Insight, Primer, Sapho, Textio, and Tule.

Machine Intelligence in the Real World

(this piece was originally posted on Tech Crunch) .

I've been laser-focused on machine intelligence in the past few years. I've talked to hundreds of entrepreneurs, researchers and investors about helping machines make us smarter.

In the months since I shared my landscape of machine intelligence companies, folks keep asking me what I think of them — as if they're all doing more or less the same thing. (I'm guessing this is how people talked about "dot coms" in 1997.)

On average, people seem most concerned about how to interact with these technologies once they are out in the wild. This post will focus on how these companies go to market, not on the methods they use.

In an attempt to explain the differences between how these companies go to market, I found myself using (admittedly colorful) nicknames. It ended up being useful, so I took a moment to spell them out in more detail so, in case you run into one or need a handy way to describe yours, you have the vernacular.

The categories aren't airtight — this is a complex space — but this framework helps our fund (which invests in companies that make work better) be more thoughtful about how we think about and interact with machine intelligence companies.

“Panopticons” Collect A Broad Dataset

Machine intelligence starts with the data computers analyze, so the companies I call “panopticons” are assembling enormous, important new datasets. Defensible businesses tend to be global in nature. “Global” is very literal in the case of a company like Planet Labs, which has satellites physically orbiting the earth. Or it's more metaphorical, in the case of a company like Premise, which is crowdsourcing data from many countries.

With many of these new datasets we can automatically get answers to questions we have struggled to answer before. There are massive barriers to entry because it's difficult to amass a global dataset of significance.

However, it's important to ask whether there is a “good enough” dataset that might provide a cheaper alternative, since data license businesses are at risk of being commoditized. Companies approaching this space should feel confident that either (1) no one else can or will collect a “good enough” alternative, or (2) they can successfully capture the intelligence layer on top of their own dataset and own the end user.

Examples include Planet Labs, Premise and Diffbot.

“Lasers” Collect A Focused Dataset

The companies I like to call “lasers” are also building new datasets, but in niches, to solve industry-specific problems with laser-like focus. Successful companies in this space provide more than just the dataset — they also must own the algorithms and user interface. They focus on narrower initial uses and must provide more value than just data to win customers.

The products immediately help users answer specific questions like, “how much should I water my crops?” or “which applicants are eligible for loans?” This category may spawn many, many companies — a hundred or more — because companies in it can produce business value right away.

With these technologies, many industries will be able to make decisions in a data-driven way for the first time. The power for good here is enormous: We’ve seen these technologies help us feed the world more efficiently, improve medical diagnostics, aid in conservation projects and provide credit to those in the world that didn’t have access to it before.

But to succeed, these companies need to find a single “killer” (meant in the benevolent way) use case to solve, and solve that problem in a way that makes the user’s life simpler, not more complex.

Examples include Tule Technologies, Enlitic, InVenture, Conservation Metrics, Red Bird, Mavrx and Watson Health.

“Alchemists” Promise To Turn Your Data Into Gold

These companies have a simple pitch: Let me work with your data, and I will return gold. Rather than creating their own datasets, they use novel algorithms to enrich and draw insights from their customers’ data. They come in three forms:

Self-service API-based solutions.

Service providers who work on top of their customers’ existing stacks.

Full-stack solutions that deliver their own hardware-optimized stacks.

Because the alchemists see across an array of data types, they’re likely to get early insight into powerful applications of machine intelligence. If they go directly to customers to solve problems in a hands-on way (i.e., with consulting services), they often become trusted partners.

But be careful. This industry is nascent, and those using an API-based approach may struggle to scale as revenue sources can only go as far as the still-small user base. Many of the self-service companies have moved toward a more hands-on model to address this problem (and those people-heavy consulting services can sometimes be harder to scale).

Examples include Nervana Systems, Context Relevant, IBM Watson, Metamind, AlchemyAPI (acquired by IBM Watson), Skymind, Lucid.ai and Citrine.

“Gateways” Create New Use Cases From Specific Data Types

These companies allow enterprises to unlock insights from a type of data they had trouble dealing with before (e.g., image, audio, video, genomic data). They don’t collect their own data, but rather work with client data and/or a third-party data provider. Unlike the Alchemists, who tend to do analysis across an array of data types and use cases, these are specialists.

What’s most exciting here is that this is genuinely new intelligence. Enterprises have generally had this data, but they either weren’t storing it or didn’t have the ability to interpret it economically. All of that “lost” data can now be used.

Still, beware the “so what” problem. Just because we have the methods to extract new insights doesn’t make them valuable. We’ve seen companies that begin with the problem they want to solve, and others blinded by the magic of the method. The latter category struggles to get funding.

Examples include Clarifai, Gridspace, Orbital Insight, Descartes Labs, Deep Genomics and Atomwise.

“Magic Wands” Seamlessly Fix A Workflow

These are SaaS tools that make work more effective, not just by extracting insights from the data you provide but by seamlessly integrating those insights into your daily workflow, creating a level of machine intelligence assistance that feels like “magic.” They are similar to the Lasers in that they have an interface that helps the user solve a specific problem — but they tend to rely on a user’s or enterprise’s data rather than creating their own new dataset from scratch.

For example, Textio is a text editor that recommends improvements to job descriptions as you type. With it, I can go from a 40th percentile job description to a 90th percentile one in just a few minutes, all thanks to a beautifully presented machine learning algorithm.

I believe that in five years we all will be using these tools across different use cases. They make the user look like an instant expert by codifying lessons found in domain-specific data. They can aggregate intelligence and silently bake it into products. We expect this space to heat up, and can’t wait to see more Magic Wands.

The risk is that by relying on such tools, humans will lose expertise (in the same way that the autopilot created the risk that pilots’ core skills may decay). To offset this, makers of these products should create UI in a way that will actually fortify the user’s knowledge rather than replace it (e.g., educating the user during the process of making a recommendation or using a double-blind interface).

Examples include Textio, RelateIQ (acquired by Salesforce), InboxVudu, Sigopt and The Grid

“Navigators” Create Autonomous Systems For The Physical World

Machine intelligence plays a huge role in enabling autonomous systems like self-driving cars, drones and robots to augment processes in warehouses, agriculture and elderly care. This category is a mix of early stage companies and large established companies like Google, Apple, Uber and Amazon.

Such technologies give us the ability to rethink transportation and logistics entirely, especially in emerging market countries that lack robust physical infrastructure. We also can use them to complete tasks that were historically very dangerous for humans.

Before committing to this kind of technology, companies should feel confident that they can raise large amounts of capital and recruit the best minds in some of the most sought-after fields. Many of these problems require experts across varied specialties, like hardware, robotics, vision and audio. They also will have to deal with steep regulatory hurdles (e.g., self-driving car regulations).

Examples include Blue River Technologies, Airware, Clearpath Robotics, Kiva Systems (acquired by Amazon), 3DR, Skycatch, Cruise Automation and the self-driving car groups at Google, Uber, Apple and Tesla.

“Agents” Create Cyborgs And Bots To Help With Virtual Tasks

Sometimes the best way to use machine intelligence is to pair it with human intelligence. Cyborgs and bots are similar in that they help you complete tasks, but the difference is a cyborg appears as if it's a human (it blends human and machine intelligence behind the scenes, has a proper name and attempts to interact like a person would), whereas a bot is explicitly non-human and relies on you to provide the human-level guidance to instruct it what to do.

Cyborgs most often complete complex tasks, like customer service via real-time chat or meeting scheduling via email (e.g., Clara from Clara Labs or Amy from x.ai). Bots tend to help you perform basic research, complete online transactions and help your team stay on top of tasks (e.g., Howdy, the project management bot).

In both cases, this is the perfect blending of humans and machines: The computers take the transactional grunt work pieces of the task and interact with us for the higher-level decision-making and creativity.

Cyborg-based companies start as mostly manual services and, over time, become more machine-driven as technology matures. The risk is whether they can make that transition quickly enough. For both cyborgs and bots, privacy and security will be an ongoing concern, as we trust more and more of our data (e.g., calendars, email, documents, credit cards) to them.

Examples include Clara, x.ai, Facebook M, Digital Genius, Kasisto and Howdy.

“Pioneers” Are Very Smart

Some machine intelligence companies begin life as academic projects. When the teams — professors and graduate students with years of experience in the field — discover they have something marketable, they (or their universities) spin them out into companies.

Aggregating a team like that is, in itself, a viable market strategy, because there are so few people with 8-10 years of experience in this field. Their brains are so valuable that investors are willing to take the risk on the basis of the team alone — even if the business models still need some work.

In fact, there are many extremely important problems to solve that don't line up with short-term use cases. These teams are the ones solving the problems that seem impossible, and they are among the few who can potentially make them possible!

This approach can work brilliantly if the team has a problem they are truly devoted to working on, but it is tough to keep the team together if they are banding together for the sake of solidarity and the prospect of an acqui-hire. They also need funders who are aligned with their longer-term vision.

Examples include DeepMind (acquired by Google), DNN Research (acquired by Google), Numenta, Vicarious, NNaiSense and Curious AI.

As you can see, it's clear that machine intelligence is a very active space. There are many companies out there that may not fit into one of these categories, but these are the ones we see most often.

The obvious question for all of these categories is which are most attractive for investment? Individual startups are outliers by definition, so it's hard to make it black and white, and we're so excited about this space that it's really just different degrees of optimism. That said, I'm particularly excited about the Lasers and Magic Wands, because they can turn new types of data into actionable intelligence right now, and because they can take advantage of well-worn SaaS techniques.

More on these to come. Stay tuned.

Disclosure: Bloomberg Beta is an investor in Diffbot, Tule Technologies, Mavrx, Gridspace, Orbital Insight, Textio, Howdy and several other machine intelligence companies that are not mentioned in this article.

The Current State of Machine Intelligence

Machine Intelligence LANDSCAPE

CORE TECHNOLOGIES

ARTIFICIAL INTELLIGENCE

IBM Watson, MetaMind, Numenta, ai-one, Cycorp, Research, nano, Reactor, SI, CALLED REFERENCE

DEEP LEARNING

vicarious, Vision Factory, facebook, Google, SKYYMIND, Baidu, ersatz, SignalSense

MACHINE LEARNING

rapidminer, context, jiji, Oxdata, H2O, DATAFARM, Liftigiter, Azure ML, yhat, turisea, Sense, GraphLab, Alpine, eonion, AYASDI

NLP PLATFORMS

cortical.io, idibon, LUMINOSO, wit.ai, Maluba

PREDICTIVE APIS

AlchemyAPI, MINDOPS, Google, big, indico, ALGORITHMIA, Expect, PredictionIO, Labs

IMAGE RECOGNITION

clarifai, MADBITS, DNNresearch, DEXTRO, VISENZE, lookflow

SPEECH RECOGNITION

GRIDSPACE, popUP archive, NUANCE

RETHINKING ENTERPRISE

SALES

Preact, AVISO, RelateIQ, NGDATA, CLARABridge, FRAMED, infer, ATTENTIVITY, causita

SECURITY / AUTHENTICATION

CROSSMATCH, c-njur, EVEREY, BITSIGHT, CYLANCE, AREA1, bionym

FRAUD DETECTION

sift science, socure, ThreatMetrix, feedzai, Brighthouse, verafin

HR / RECRUITING

TalentBin, entelo, predict, Connectifier, gild, hiQ, CONCEPT

MARKETING

brightfunnel, bloomreach, CommandIQ, AIRPR, RADIUS, Telapart, people, PATTERN, Freemium

PERSONAL ASSISTANT

Siri, Google now, Cortana, cleversense, tempo, Rebinlabs, KASIST, fuse/machines, VIV, CLARA LABS

INTELLIGENCE TOOLS

ADATAD, Palantir, Quid, Digital Reasoning, FirstRain

RETHINKING INDUSTRIES

ADTECH

METAMARKETS, dstillery, rocketfuel, YieldMo, ADBRAIN

AGRICULTURE

BLUEGRIVER, Terraviva, cereals, HONDA, THE CLIMATE CORPORATION, tule, XEROX

EDUCATION

edclara, coursera, KNEWTON, kidaptive

FINANCE

Bloomberg, alphasense, KENSHC, minetabrook, Datominr, BINATIX

LEGAL

Lex Machina, brightleaf, COUNSELYTICS, RAVEL, JUDICATA, Brevia, DiligenceEngine

MANUFACTURING

SIGHT MACHINE, MICROSCAN, IVISYS, BINDER IMAGING

MEDICAL

Parzival, transcriptic, ZEPHYR, Genescent, bino, TUTE, grand-mound, bator

OIL AND GAS

kaggle, biota, TACHYUS, Futura

MEDIA / CONTENT

Outbrain, newstle, ARRIA, SAILTHRU, wovii, NarrativeScience, SCOP, Summary, Prismatic, ai, AUTOMATED

CONSUMER FINANCE

Affirm, inVenture, finance, BULLY GUARD, LendUp, LendingClub, Kabbage

PHILANTHROPIES

DataKind, thorn, DATA GUILD

AUTOMOTIVE

Google, Continental, T, Cruise, Cruise

DIAGNOSTICS

enlithic, 3SCAN, lumia, ENTOTIS

RETAIL

BAY SENSORS, PRISM SKYLABS, celect, euclid

RETHINKING HUMANS / HCI

AUGMENTED REALITY

Microsoft, APX, blippar, META, layer

GESTURAL COMPUTING

THALMICLABS, omek, THALMICLABS, LEAP, 3Gear, Gesturetek, nod

ROBOTICS

intel, LIQUID ROBOTICS, iRobot, Boston Dynamics, QRL

EMOTIONAL RECOGNITION

affectiva, BEYONDVERBAL, EMOTIENT, cogito

SUPPORTING TECHNOLOGIES

HARDWARE

NVIDIA, XILINX, QUALCOMM, NERVANA, TONKIEP, ARTIFICIAL LEARNING, rigetti

DATA PREP

TRIFACTA, Paxata, tamr, Alation

DATA COLLECTION

diffbot, kimono, CrowdFlower, Cnnnotate, Workfusion, Import

www.shivonzilis.com/machineintelligence

Bloomberg BETA

I spent the last three months learning about every artificial intelligence, machine learning, or data related startup I could find — my current list has 2,529 of them to be exact. Yes, I should find better things to do with my evenings and weekends but until then...

Why do this?

A few years ago, investors and startups were chasing “big data” (I helped put together a landscape on that industry). Now we’re seeing a similar explosion of companies calling themselves artificial intelligence, machine learning, or somesuch — collectively I call these “machine intelligence” (I’ll get into the definitions in a second). Our fund, Bloomberg Beta, which is focused on the future of work, has been investing in these approaches. I created this landscape to start to put startups into context. I’m a thesis-oriented investor and it’s much easier to identify crowded areas and see white space once the landscape has some sort of taxonomy.

What is “machine intelligence,” anyway?

I mean “machine intelligence” as a unifying term for what others call machine learning and artificial intelligence. (Some others have used the term before, without quite describing it or understanding how laden this field has been with debates over descriptions.)

I would have preferred to avoid a different label but when I tried either “artificial intelligence” or “machine learning” both proved to be too narrow: when I called it “artificial intelligence” too many people were distracted by whether certain companies were “true AI,” and when I called it “machine learning,” many thought I wasn’t doing justice to the more “AI-esque” like the various flavors of deep learning. People have immediately grasped “machine intelligence” so here we are. ☺ Computers are learning to think, read, and write. They’re also picking up human sensory function, with the ability to see and hear (arguably to touch, taste, and smell, though those have been of a lesser focus). Machine intelligence technologies cut across a vast array of problem types (from classification and clustering to natural language processing and computer vision) and methods (from support vector machines to deep belief networks). All of these technologies are reflected on this landscape.

What this landscape doesn’t include, however important, is “big data” technologies. Some have used this term interchangeably with machine learning and artificial intelligence, but I want to focus on the intelligence methods rather than data, storage, and computation pieces of the puzzle for this landscape (though of course data technologies enable machine intelligence).

Which companies are on the landscape?

I considered thousands of companies, so while the chart is crowded it’s still a small subset of the overall ecosystem. “Admissions rates” to the chart were fairly in line with those of Yale or Harvard, and perhaps equally arbitrary. ☺

I tried to pick companies that used machine intelligence methods as a defining part of their technology. Many of these companies clearly belong in multiple areas but for the sake of simplicity I tried to keep companies in their primary area and categorized them by the language they use to describe themselves (instead of quibbling over whether a company used “NLP” accurately in its self-description).

If you want to get a sense for innovations at the heart of machine intelligence, focus on the core technologies layer. Some of these companies have APIs that power other applications, some sell their platforms directly into enterprise, some are at the stage of cryptic demos, and some are so stealthy that all we have is a few

sentences to describe them.

The most exciting part for me was seeing how much is happening in the application space. These companies separated nicely into those that reinvent the enterprise, industries, and ourselves.

If I were looking to build a company right now, I'd use this landscape to help figure out what core and supporting technologies I could package into a novel industry application. Everyone likes solving the sexy problems but there are an incredible amount of 'unsexy' industry use cases that have massive market opportunities and powerful enabling technologies that are begging to be used for creative applications (e.g., Watson Developer Cloud, AlchemyAPI).

Reflections on the landscape:

We've seen a few great articles recently outlining why machine intelligence is experiencing a resurgence, documenting the enabling factors of this resurgence. (Kevin Kelly, for example chalks it up to cheap parallel computing, large datasets, and better algorithms.) I focused on understanding the ecosystem on a company-by-company level and drawing implications from that.

Yes, it's true, machine intelligence is transforming the enterprise, industries and humans alike.

On a high level it's easy to understand why machine intelligence is important, but it wasn't until I laid out what many of these companies are actually doing that I started to grok how much it is already transforming everything around us. As Kevin Kelly more provocatively put it, "the business plans of the next 10,000 startups are easy to forecast: Take X and add AI". In many cases you don't even need the X — machine intelligence will certainly transform existing industries, but will also likely create entirely new ones.

Machine intelligence is enabling applications we already expect like automated assistants (Siri), adorable robots (Jibo), and identifying people in images (like the highly effective but unfortunately named DeepFace). However, it's also doing the unexpected: protecting children from sex trafficking, reducing the chemical content in the lettuce we eat, helping us buy shoes online that fit our feet precisely, and destroying 80's classic video games.

Many companies will be acquired.

I was surprised to find that over 10% of the eligible (non-public) companies on the slide have been acquired. It was in stark contrast to big data landscape we created, which had very few acquisitions at the time. No jaw will drop when I reveal that Google is the number one acquirer, though there were more than 15 different acquirers just for the companies on this chart. My guess is that by the end of 2015 almost another 10% will be acquired. For thoughts on which specific ones will get snapped up in the next year you'll have to twist my arm...

Big companies have a disproportionate advantage, especially those that build consumer products.

The giants in search (Google, Baidu), social networks (Facebook, LinkedIn, Pinterest), content (Netflix, Yahoo!), mobile (Apple) and e-commerce (Amazon) are in an incredible position. They have massive datasets and constant consumer interactions that enable tight feedback loops for their algorithms (and these factors combine to create powerful network effects) — and they have the most to gain from the low hanging fruit that machine intelligence bears.

Best-in-class personalization and recommendation algorithms have enabled these companies' success (it's both impressive and disconcerting that Facebook recommends you add the person you had a crush on in college and Netflix tees up that perfect guilty pleasure sitcom). Now they are all competing in a new battlefield: the move to mobile. Winning mobile will require lots of machine intelligence: state of the art natural language interfaces (like Apple's Siri), visual search (like Amazon's "FireFly"), and dynamic question answering technology that tells you the answer instead of providing a menu of links (all of the search companies are wrestling with this). Large enterprise companies (IBM and Microsoft) have also made incredible strides in the field, though they don't have the same human-facing requirements so are focusing their attention more on knowledge representation tasks on large industry datasets, like IBM Watson's application to assist doctors with diagnoses.

The talent is in the New AI Ivy League.

In the last 20 years, most of the best minds in machine intelligence (especially the 'hardcore AI' types) worked in academia. They developed new machine intelligence methods, but there were few real world applications that could drive business value.

Now that real world applications of more complex machine intelligence methods like deep belief nets and hierarchical neural networks are starting to solve real world problems, we're seeing academic talent move to corporate settings. Facebook recruited NYU professors Yann LeCun and Rob Fergus to their AI Lab, Google hired University of Toronto's Geoffrey Hinton, Baidu wooed Andrew Ng. It's important to note that they all still give back significantly to the academic community (one of LeCun's lab mandates is to work on core research to give back to the community, Hinton spends half of his time teaching, Ng has made machine intelligence more accessible through Coursera) but it is clear that a lot of the intellectual horsepower is moving away from academia.

For aspiring minds in the space, these corporate labs not only offer lucrative salaries and access to the "godfathers" of the industry, but, the most important ingredient: data. These labs offer talent access to datasets they could never get otherwise (the ImageNet dataset is fantastic, but can't compare to what Facebook, Google, and Baidu have in house).

As a result, we'll likely see corporations become the home of many of the most important innovations in machine intelligence and recruit many of the graduate students and postdocs that would have otherwise stayed in academia.

There will be a peace dividend.

Big companies have an inherent advantage and it's likely that the ones who will win the machine intelligence race will be even more powerful than they are today. However, the good news for the rest of the world is that the core technology they develop will rapidly spill into other areas, both via departing talent and published research.

Similar to the big data revolution, which was sparked by the release of Google's BigTable and BigQuery papers, we will see corporations release equally groundbreaking new technologies into the community. Those innovations will be adapted to new industries and use cases that the Googles of the world don't have the DNA or desire to tackle.

Opportunities for entrepreneurs:

"My company does deep learning for X"

Few words will make you more popular in 2015. That is, if you can credibly say them. Deep learning is a particularly popular method in the machine intelligence field that has been getting a lot of attention. Google, Facebook, and Baidu have achieved excellent results with the method for vision and language based tasks and startups like Enlitic have shown promising results as well.

Yes, it will be an overused buzzword with excitement ahead of results and business models, but unlike the hundreds of companies that say they do "big data", it's much easier to cut to the chase in terms of verifying credibility here if you're paying attention. The most exciting part about the deep learning method is that when applied with the appropriate levels of care and feeding, it can replace some of the intuition that comes from domain expertise with automatically-learned features. The hope is that, in many cases, it will allow us to fundamentally rethink what a best-in-class solution is.

As an investor who is curious about the quirkier applications of data and machine intelligence, I can't wait to see what creative problems deep learning practitioners try to solve. I completely agree with Jeff Hawkins when he says a lot of the killer applications of these types of technologies will sneak up on us. I fully intend to keep an open mind.

"Acquire as a business model"

People say that data scientists are unicorns in short supply. The talent crunch in machine intelligence will make it look like we had a glut of data scientists. In the data field, many people had industry experience over the past decade. Most hardcore machine intelligence work has only been in academia. We won't be able to grow this talent overnight.

This shortage of talent is a boon for founders who actually understand machine intelligence. A lot of companies in the space will get seed funding because there are early signs that the acquihire price for a machine intelligence expert is north of 5x that of a normal technical acquihire (take, for example Deep Mind, where price per technical head was somewhere between \$5–10M, if we choose to consider it in the acquihire category). I've had multiple friends ask me, only semi-jokingly, "Shivon, should I just round up all of my smartest friends in the AI world and call it a company?" To be honest, I'm not sure what to tell them. (At Bloomberg Beta, we'd rather back companies building for the long term, but that doesn't mean this won't be a lucrative strategy for many enterprising founders.)

A good demo is disproportionately valuable in machine intelligence

I remember watching Watson play Jeopardy. When it struggled at the beginning I felt really sad for it. When it started trouncing its competitors I remember cheering it on as if it were the Toronto Maple Leafs in the Stanley Cup finals (disclaimers: (1) I was an IBMer at the time so was biased towards my team (2) the Maple Leafs have not made the finals during my lifetime — yet — so that was purely a hypothetical).

Why do these awe-inspiring demos matter? The last wave of technology companies to IPO didn't have demos that most of us would watch, so why should machine intelligence companies? The last wave of companies were very computer-like: database companies, enterprise applications, and the like. Sure, I'd like to see a 10x more performant database, but most people wouldn't care. Machine intelligence wins and loses on demos because 1) the technology is very human, enough to inspire shock and awe, 2) business models tend to take a while to form, so they need more funding for longer period of time to get them there, 3) they are fantastic acquisition bait. Watson beat the world's best humans at trivia, even if it thought Toronto was a US city. DeepMind blew people away by beating video games. Vicarious took on CAPTCHA. There are a few companies still in stealth that promise to impress beyond that, and I can't wait to see if they get there.

Demo or not, I'd love to talk to anyone using machine intelligence to change the world. There's no industry too unsexy, no problem too geeky. I'd love to be there to help so don't be shy. I hope this landscape chart sparks a conversation. The goal is to make this a living document and I want to know if there are companies or categories missing. I welcome feedback and would like to put together a dynamic visualization where I can add more companies and dimensions to the data (methods used, data types, end users, investment to date, location, etc.) so that folks can interact with it to better explore the space.

Disclaimer: Bloomberg Beta is an investor in Adatao, Alation, Aviso, Context Relevant, Mavrx, Newsle, Orbital Insights, Pop Up Archive, and two others on the chart that are still undisclosed. We're also investors in a few other machine intelligence companies that aren't focusing on areas that were a fit for this landscape, so we left them off.

For the full resolution version of the landscape please [click here](#).

Copyright ©2025

